

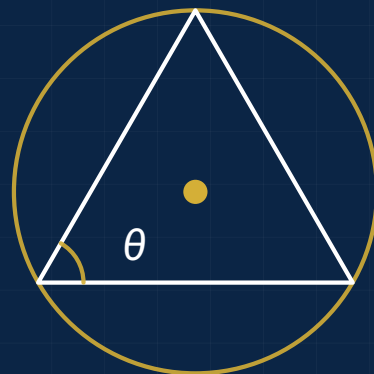


DR ESLAM AHMED
eliteigcse.com

CHAPTER 4 • GEOMETRY & MEASURES • TOPIC 7

Volume & Surface Area

Prism: cross-section $\times$ length. Cone and pyramid: a third of that. Surface area: add the area of every face you could paint.



4

VARIANTS

22

PAST-PAPER PARTS

65

PRACTICE QUESTIONS

YOU WILL MASTER

V1 Prisms & cylinders

V2 Cones, pyramids & spheres

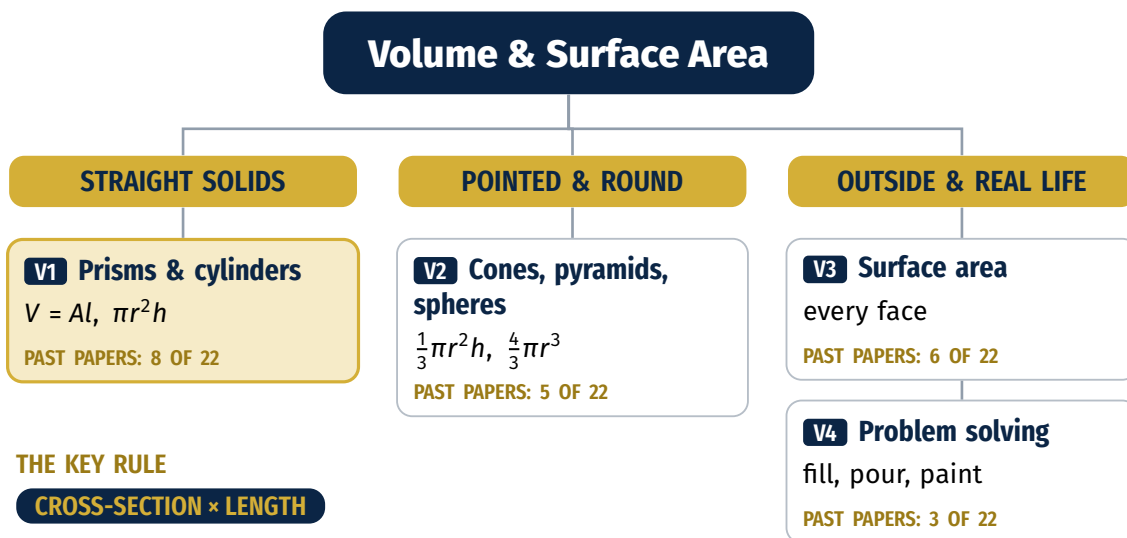
V3 Surface area

V4 Problem solving

TOPIC MAP • EVERY VARIANT ON ONE PAGE

Volume at a glance



Your notes


THE KEY RULE

CROSS-SECTION × LENGTH

Any prism, any shape of end.

★ How often does each variant come up?

20 past-paper questions (Jan 2020 → Jun 2025) hold 22 parts; 65 practice parts — grouped by type.

	PAST PAPERS (OUT OF 22)	PRACTICE PARTS
V1 Prisms, cylinders	8	17
V2 Cones, pyramids, spheres	5	15
V3 Surface area	6	15
V4 Problem solving	3	18

WHAT THIS TELLS YOU On every paper: an easy prism/cylinder (3 marks, early) and a harder sphere/cone question (5–6 marks, near the end). The formulas for cones and spheres are on the formula sheet — know when to use them.

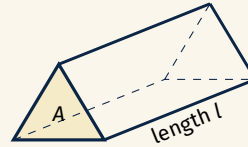
BEFORE YOU START

The big idea

A prism is the same all the way through.

Slice it anywhere: the same **cross-section**. So volume = **area of the end** × **length**.

Pointed solids (cone, pyramid) hold exactly **a third** of the matching prism.



$$V = A \times l$$

 Your notes

The formulas (cone and sphere are on the formula sheet)

Cylinder

$$\pi r^2 h$$

Curved area $2\pi rh$.

Cone

$$\frac{1}{3}\pi r^2 h$$

Curved area πrl , l = slant.

Sphere

$$\frac{4}{3}\pi r^3$$

Surface $4\pi r^2$.

Pyramid

$$\frac{1}{3} \times \text{base} \times h$$

h = vertical height.

Units

Litres

1 litre = 1000 cm^3
 $\div 1000$.

m^3 to cm^3

$\times 1\,000\,000$
 100^3 .

Hemisphere

$$\frac{2}{3}\pi r^3$$

Surface: $2\pi r^2$ curved + πr^2 flat.



CALCULATOR FIRST, THEN CHECK

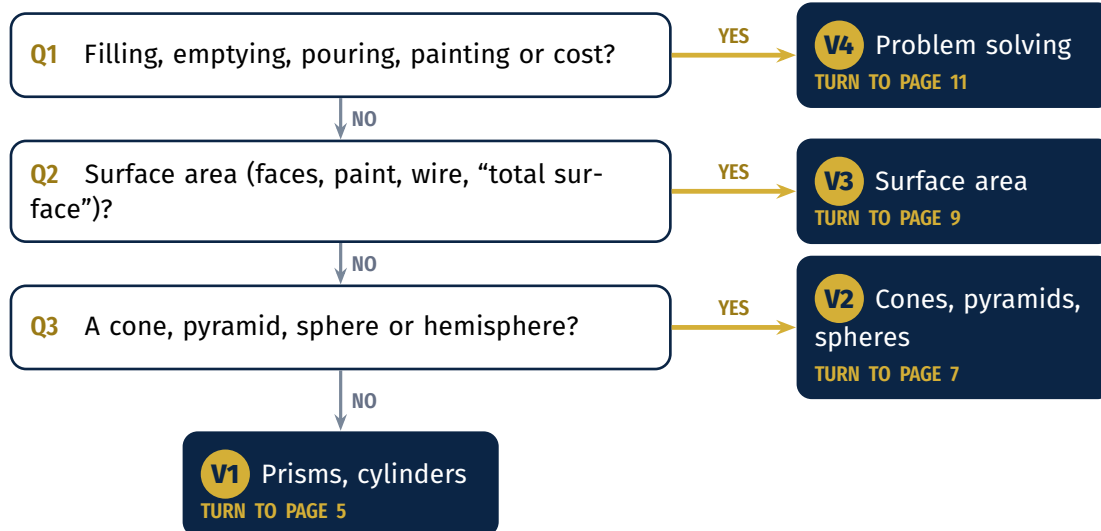
Exact answers: keep π (666π). Rounded answers: full calculator value, round once.

COUNT / RECOGNISE, THEN CHOOSE

Which variant am I?

 Your notes

Read the question, then walk down the ladder. The first **YES** tells you the variant — and the page to turn to.



COLOUR CODE IN THE WORKING

Blue = letter terms $4x$, $-3y$

Orange = plain numbers 17 , -13

STEP 1 **STEP 2** **STEP 3** **STEP 4** = which step of "How do I do it?" each line is.

ON EVERY VARIANT SPREAD

- | | |
|---|---|
|  What is it? |  When you see → you do |
|  How do I do it? |  Watch out |
|  Worked example |  Exam tip |
|  Check yourself |  Calculator check |

V1

VARIANT 1 MOST COMMON IN THE EXAM

Prisms & Cylinders

Area of the cross-section $\times$ length; cylinder $\pi r^2 h$. Backwards: divide.

SEEN IN PAST PAPERS

8 / 22

PRACTICE QUESTIONS: 17


 Your notes

Came up in: Jan 2020 P2HR Q13 • Jan 2021 P1HR Q5 • May 2021 P2H Q10 • Jun 2023 P2HR Q9 • Nov 2023 P1H Q21 • Nov 2024 P1H Q5 • Jun 2025 P1H Q7

? What is it?

In plain words

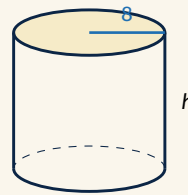
Find the **area of the end face** (triangle, trapezium, circle ...), then multiply by the **length**.

Volume given? **Divide** to get the missing length: $h = \frac{V}{\pi r^2}$.

IT LOOKS LIKE

trapezium prism

$$\pi r^2 h = 3892$$



$$\pi(8)^2 h = 3892$$

How do I do it?

1 End face

Name it; find its area.

2 Length

The distance between the two ends.

3 Multiply

Area $\times$ length (cylinder: $\pi r^2 h$).

4 Backwards

Volume given: divide.

CROSS-SECTION $\times$ LENGTH The most common variant here.

V1 Prisms & Cylinders

VARIANT 1 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

"The volume of the cylinder is 3892 cm³. Work out h "

$$\rightarrow h = 3892 \div (\pi \times 8^2)$$

"Give your answer in litres"

$$\rightarrow \text{cm}^3 \div 1000$$

A triangle on the end with an angle

$$\rightarrow \frac{1}{2}ab \sin C \text{ for the cross-section}$$

WORKED EXAMPLE **PAST PAPER**

June 2025 · Paper 1H · Q7 · 3 marks

A solid cylinder has radius 8 cm and height h cm. Its volume is 3892 cm³. Work out h (1 d.p.).

STEP 1 $V = \pi r^2 h$ ◀ cylinder

STEP 2 $\pi \times 64 \times h = 3892$ ◀ substitute

STEP 3 $h = \frac{3892}{64\pi} = 19.357 \dots$ ◀ divide

STEP 4 ≈ 19.4 ◀ 1 d.p.

ANSWER $h = 19.4$

$$\checkmark \pi \times 64 \times 19.4 \approx 3901 \approx 3892 \checkmark$$

WATCH OUT

$$\times \pi \times 8 \times h$$

$$\checkmark \pi \times 8^2 \times h$$

Square the radius.

EXAM TIP

Write the formula with the numbers in before you rearrange.

Check yourself

calculator first · answers on page 13

1.1

Dr Eslam

A prism has cross-section 18 cm² and length 12 cm. Volume?

Answer:

1.2

Nov 2024 P1H

A cylinder has radius 70 cm and height 18 cm. Volume in litres (nearest litre)?

Answer:

1.3

Jan 2020 P2HR

A prism has a trapezium cross-section (7 and 12 cm, height h), length 10 cm, volume 608 cm³. Find h .

Answer:

V2

VARIANT 2

Cones, Pyramids & Spheres

Use the formula sheet: cone $\frac{1}{3}\pi r^2 h$, sphere $\frac{4}{3}\pi r^3$; hemisphere is half.

SEEN IN PAST PAPERS

5 / 22

PRACTICE QUESTIONS: 15

 Your notes

Came up in: Jan 2020 P2H Q26 • Jun 2022 P1H Q22 • Jun 2023 P2HR Q24 • May 2024 P1H Q23 • Nov 2024 P1H Q22

? What is it?

In plain words

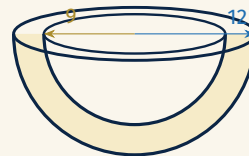
These formulas are **given** on the formula sheet. The skill is choosing the right one and the right **radius**.

A **hemisphere** is half a sphere. A **hollow** bowl = outer hemisphere - inner hemisphere.

IT LOOKS LIKE

bowl: outer 12

inner 9



$$\frac{2}{3}\pi(12^3 - 9^3)$$

How do I do it?

1 Pick

Cone, pyramid, sphere, hemisphere?

2 Radius

Halve a diameter; check which radius.

3 FormulaFrom the sheet; hemisphere $\frac{2}{3}\pi r^3$.**4** Combine

Add or subtract solids.

1/3 FOR POINTED SOLIDSHemisphere: half of $\frac{4}{3}$.

V2 Cones, Pyramids & Spheres

VARIANT 2 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

"bowl ... outer radius 12 cm, inner radius 9 cm"

→ **Outer – inner hemisphere**

"surface area of A is nine times that of B"

→ **Radius ×3, volume ×27**

"exact value of k" with $k\pi$

→ **Leave π out of the arithmetic****WORKED EXAMPLE****PAST PAPER**

November 2024 · Paper 1H · Q22 · 3 marks

A hemispherical metal bowl has outer radius 12 cm and inner radius 9 cm. Work out the volume of the metal (nearest whole number).

STEP 1 $\frac{2}{3}\pi(12)^3 = 1152\pi$ ◀ outer hemisphere

STEP 2 $\frac{2}{3}\pi(9)^3 = 486\pi$ ◀ inner hemisphere

STEP 3 $1152\pi - 486\pi = 666\pi$ ◀ subtract

STEP 4 $= 2092.3 \dots$ ◀

ANSWER 2092 cm³

$$\checkmark \frac{2}{3}\pi(1728 - 729) = 666\pi \checkmark$$

**WATCH OUT**

$$\times \frac{2}{3}\pi(12 - 9)^3$$

$$\checkmark \frac{2}{3}\pi(12^3 - 9^3)$$

Cube each radius, then subtract.

**EXAM TIP**

Write everything as a multiple of π until the last line.

**Check yourself**

calculator first · answers on page 13

2.1

Dr Eslam

A cone has radius 6 cm and height 10 cm. Volume (3 s.f.)?

Answer:

2.2

Jun 2023 P2HR

Sphere A has 9 times the surface area of sphere B; the volumes differ by 117π cm³. Radius of B?

Answer:

2.3

Jan 2020 P2H

A sector of angle x° makes a cone of height 25 cm and volume 1600 cm³. Find x .

Answer:

V3

VARIANT 3

Surface Area

List every face, find each area, add. Curved faces: $2\pi rh$, πrl , $4\pi r^2$.

SEEN IN PAST PAPERS

6 / 22

PRACTICE QUESTIONS: 15



Your notes

Came up in: Jan 2021 P2H Q17 • Nov 2021 P2H Q22 • Nov 2024 P2H Q6 • Jun 2025 P2HR Q26 • Jun 2025 P2H Q21

? What is it?**In plain words**

Surface area is the area of **every face**

— like the paper needed to wrap it.

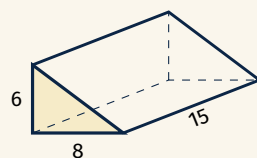
Count the faces: a triangular prism has **2 triangles + 3 rectangles**.

Missing lengths (like the hypotenuse) come from **Pythagoras**.

IT LOOKS LIKE

triangular prism

hemisphere bowl



$$2 \times 24 + 15(6 + 8 + 10)$$

How do I do it?**1 Faces**

List them all (sketch a net).

2 Lengths

Pythagoras for slants and hypotenuses.

3 Areas

Each face, curved parts by formula.

4 Add

Total; “show that”: every stage.

EVERY FACE, ONCE Hemisphere: curved + flat circle.

V3 Surface Area

VARIANT 3 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“Work out the total surface area of the triangular prism” → **2 triangles + 3 rectangles**

“Show that $A = 12x^2 + \frac{1690}{x}$ ” → **y from the volume, then 6 faces**

“total surface area of the bowl ... in terms of π ” → **Outer + inner curved + the rim ring**

WORKED EXAMPLE **PAST PAPER**

November 2024 · Paper 2H · Q6 · 3 marks

A solid triangular prism has a right-angled triangle cross-section with sides 6 cm and 8 cm, and length 15 cm. Work out the total surface area.

STEP 1 $\sqrt{6^2 + 8^2} = 10$ ◀ third side

STEP 2 $2 \times \frac{1}{2} \times 6 \times 8 = 48$ ◀ two triangles

STEP 3 $15 \times (6 + 8 + 10) = 360$ ◀ three rectangles

STEP 4 $48 + 360 = 408$ ◀ add

ANSWER 408 cm²

✓ 5 faces counted ✓

WATCH OUT

- ✗ one triangle
- ✓ two triangles (both ends)

A prism has two end faces.

EXAM TIP

The rectangles together are length × perimeter of the end: 15×24 .

Check yourself

calculator first · answers on page 13

3.1

Dr Eslam

Total surface area of a solid hemisphere of radius 4 cm, in terms of π ?

Answer:

3.2

Jun 2025 P2H

A cuboid $3x$ by $2x$ by y has volume 1014 cm^3 . Show the total surface area $A = 12x^2 + \frac{1690}{x}$.

Answer:

3.3

Jun 2025 P2HR

A hemisphere of volume $6174\pi \text{ cm}^3$ is hollowed to a bowl 2 cm thick. Total surface area of the bowl?

Answer:

V4

VARIANT 4

Problem Solving with Volume

Turn the story into volumes (or areas), compare or divide, then answer the question in words.

SEEN IN PAST PAPERS

3 / 22

PRACTICE QUESTIONS: 18

 Your notes

Came up in: Jan 2021 P1H Q3 • Jan 2022 P1HR Q10 • Jun 2025 P1HR Q10

? What is it?

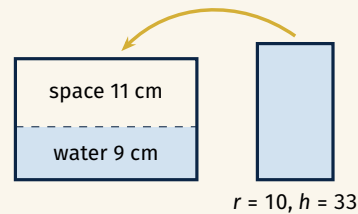
In plain words

Water poured from one container into another: the **volume stays the same**. Compare it with the **space** available.

Rates (cm^3 per minute), tins of paint, costs: one step at a time, with a **conclusion**.

IT LOOKS LIKE

pour the cylinder into the cuboid



How do I do it?

1 Volumes

Each container / the water / the space.

2 Same units

cm^3 , litres, m^3 .

3 Compare or divide

Fits? How many? How long?

4 Conclude

A sentence with the numbers.

VOLUME IS KEPT Compare with the space left.

V4 Problem Solving with Volume

VARIANT 4 · SEE IT DONE, THEN TRY



Your notes
WHEN YOU SEE → YOU DO

“Show that the cuboid will not be completely full”

→ **Water volume < space left**

“Find the time at which all the water has been pumped out”

→ **Rate from one stage, then total time**

**WORKED EXAMPLE** **PAST PAPER**

June 2025 · Paper 1HR · Q10 · 3 marks

A cuboid 35 cm by 28 cm by 20 cm holds water 9 cm deep. A full cylinder (radius 10 cm, height 33 cm) is poured in. Show that the cuboid will not be full.

STEP 1 $\pi \times 10^2 \times 33 = 3300\pi = 10\,367$

◀ water in the cylinder

STEP 2+3 $35 \times 28 \times (20 - 9) = 10\,780$

◀ space above the water

STEP 4 $10\,367 < 10\,780$

◀ compare

ANSWER it fits — the cuboid is not completely full ✓ 413 cm³ left over ✓

**WATCH OUT**

✗ compare with the whole cuboid (19 600)

✓ compare with the space left

9 cm of water is already in it.

**EXAM TIP**

“Show that”: both numbers and the comparison sentence are needed.

**Check yourself**

calculator first · answers on page 13

4.1

Jan 2022 P1HR

Paint 4 sides and top of a 12 m × 2.5 m × 3 m box; tins cover 15 m²; £26.95 a tin after a 10% rise. Cost before?

Answer:

4.2

Jan 2021 P1H

500 000 cm³ of water; 75 000 cm³ pumped out from 10:30 to 12:00, steadily. When is it empty?

Answer:

4.3

Practice · Medium

12 hemispherical bowls of radius 20 cm are filled from a 200-litre tank. Is there enough water?

Answer:

CHECK YOURSELF

Answers



Every answer below was checked with sympy (check_answers.py).

1.1 216 cm^3

1.2 277 litres

1.3 $h = 6.4$

2.1 377 cm^3

2.2 1.5 cm

2.3 $x = 107$

3.1 $48\pi \text{ cm}^2$

3.2 $A = 12x^2 + \frac{1690}{x}$

3.3 $1684\pi \text{ cm}^2$

4.1 $\text{£}196 < \text{£}200$

4.2 20:30

4.3 No: 201 litres needed

THE WHOLE TOPIC ON ONE PAGE

Volume cheat sheet



Your notes
V1 Prisms 8/22

$$A \times l, \pi r^2 h$$

Backwards: divide.

V2 Pointed, round 5/22

$$\frac{1}{3}\pi r^2 h, \frac{4}{3}\pi r^3$$

Formula sheet; hemisphere half.

V3 Surface 6/22

$$2 \times 24 + 15 \times 24$$

Every face once.

V4 Problems 3/22

$$10\,367 < 10\,780$$

Volume kept; conclude.

**TOP 3 LOST MARKS**

1. r not squared or cubed $\times$
2. one end face missing $\times$
3. no conclusion in “show that” $\times$

BOTTOM LINE Cross-section $\times$ length · a third for points · every face once · 1 litre = 1000 cm³.